

MUHAMMAD UMER BAPPI

Computer Science Graduate | AI / ML

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PROFESSIONAL SUMMARY

Computer Science graduate focused on Machine Learning, NLP, and applied AI, with hands-on experience designing and shipping end-to-end AI systems. Skilled in LLM fine-tuning (LoRA/QLoRA), retrieval-augmented generation, and multi-agent architectures, including a framework that reduced hallucinations by 35% in Urdu NLP. Comfortable across the full AI lifecycle, from data and fine-tuning through deployment and evaluation.

PROFESSIONAL EXPERIENCE

Inteligo

April 2026 – Present

Junior GHL Associate

- Design and build automated client workflows on the GoHighLevel (GHL) platform, integrating CRMs, webhooks, and third-party APIs to streamline lead capture, customer communications, and operational pipelines.
- Translate client business requirements into technical workflow specifications, architecting conditional logic, multi-step automations, and API integrations across SaaS tools.
- Debug, optimise, and ship production automations end-to-end, applying software engineering practices such as version control, modular design, and systematic testing to ensure reliable client delivery.

Revotic AI — Remote

September 2025 – November 2025

ML/AI Intern

- Designed and built an ML-based weather forecast system end-to-end — data preprocessing, feature engineering, model training, and evaluation — delivering production-ready predictions.
- Shipped production AI features in a fast-paced startup, collaborating directly with senior engineers on applied ML problems.
- Translated business requirements into technical ML specifications, bridging the gap between stakeholder needs and model design.

Forman Christian College (University) — Lahore, Pakistan

February 2024 – December 2025

Research Assistant

- Identified research gaps in ML/NLP and independently designed experimental frameworks, applying state-of-the-art methodologies to validate novel hypotheses.
- Co-authored research paper on LLM hallucination reduction in Urdu NLP, submitted to international conferences; contributed to manuscript preparation and experimental design.

EDUCATION

Forman Christian College (University), Pakistan

December 2025

Bachelor of Science in Computer Science

Key Courses: Data Structures & Algorithms, Operating Systems, Databases, Networks, AI, Machine Learning, NLP, Game Development, Information Security, Discrete Mathematics, Java.

TECHNICAL SKILLS

Languages & Frameworks: Java, Python, C++, React, Node.js, FastAPI

Databases: PostgreSQL, MySQL

DevOps & Cloud: Docker, GCP, CI/CD

Tools & Methodologies: Git, Confluence, REST APIs, Claude Code

Automation: n8n, Zapier, Make, GoHighLevel (GHL)

ML / DL Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, PEFT, Unsloth, MLX

LLM & NLP: LLM Fine-Tuning (LoRA, QLoRA), Prompt Engineering, RAG, Agentic Architectures, Text Generation, LangChain, FAISS

Evaluation & Data: BERTScore, Exact Match, Data Augmentation, Multi-format Processing (JSON/JSONL/CSV), Pandas, NumPy

PROJECTS

- Dental Voice Agent — AI Voice Receptionist**
Developed an AI-powered voice receptionist for dental clinics using Vapi, GPT-4o, and FastAPI. The system handles live phone calls autonomously — booking, rescheduling, and cancelling appointments via Google Calendar integration, answering clinic enquiries, and providing real-time availability checks across multiple dentists.
- U-CoVe — Multi-Agent Hallucination Mitigation for Urdu LLMs (Final Year Project)**
Built U-CoVe, a multi-agent verification framework that reduced hallucination rates in Urdu LLMs (Alif 1.0 and Lughaat) by 35% (relative). The framework externalises verification to two specialised fine-tuned agents: an mBERT hallucination classifier and an mT5 extractive QA agent, trained on synthetic data generated with GPT-4o-mini. Also released Halu-UQA, the first hallucination benchmark for Urdu (10,000 expert-validated QA pairs). Paper submitted to LREC 2026.
- Lead Management Automation — n8n**
Built an automated lead qualification pipeline using n8n, OpenAI GPT-4o-mini, and Google Sheets. Typeform submissions are AI-scored by budget and urgency into Hot/Warm/Cold tiers, with hot leads receiving automated personalised follow-up emails via Gmail and real-time Slack notifications for all inbound leads.
- LyricGen — AI-Powered Lyric Generation from Audio Analysis**
Fine-tuned Llama 3.2-3B with LoRA (Unsloth, 4-bit quantization) on a curated 605-pair subset of the Million Song Dataset to generate lyrics conditioned on audio features extracted with Librosa (tempo, key, mode, loudness, time signature). Built an interactive Gradio web app with chat-based lyric refinement; model published to Hugging Face Hub.
- FocusScaffold — AI-Powered ADHD Productivity Web Application**
Built a full-stack ADHD productivity tool featuring AI-driven task decomposition, cognitive dump workflows, focus session tracking, and a weekly pattern dashboard, designed to reduce cognitive load for users with executive function challenges.
- Real-Time Face Detection & Age Estimation System**
Built a real-time face detection and age estimation pipeline using OpenCV, DeepFace, and a Random Forest classifier, processing live video streams for instant predictions.
- Custom Tower Defence Game — Godot**
Developed an original tower defence game in Godot featuring a custom story plot, original scenes, and full gameplay mechanics from scratch.

AWARDS & HONORS

- Best Project Award — COMSPEX** 2025
Forman Christian College (University)
Recognised for research project on LLM hallucination in Urdu NLP.

EXTRACURRICULAR ACTIVITIES

- Music:** Session Lead Guitarist & Music Producer | 1st Place, Battle of Bands (CMH Lahore 2024) | Music Coordinator, Forman Music Society (2022–2024)
- Sports:** Football, Swimming, Table Tennis
- Reading:** Technical books and blogs